

Subnetting

4

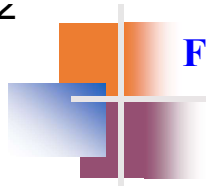
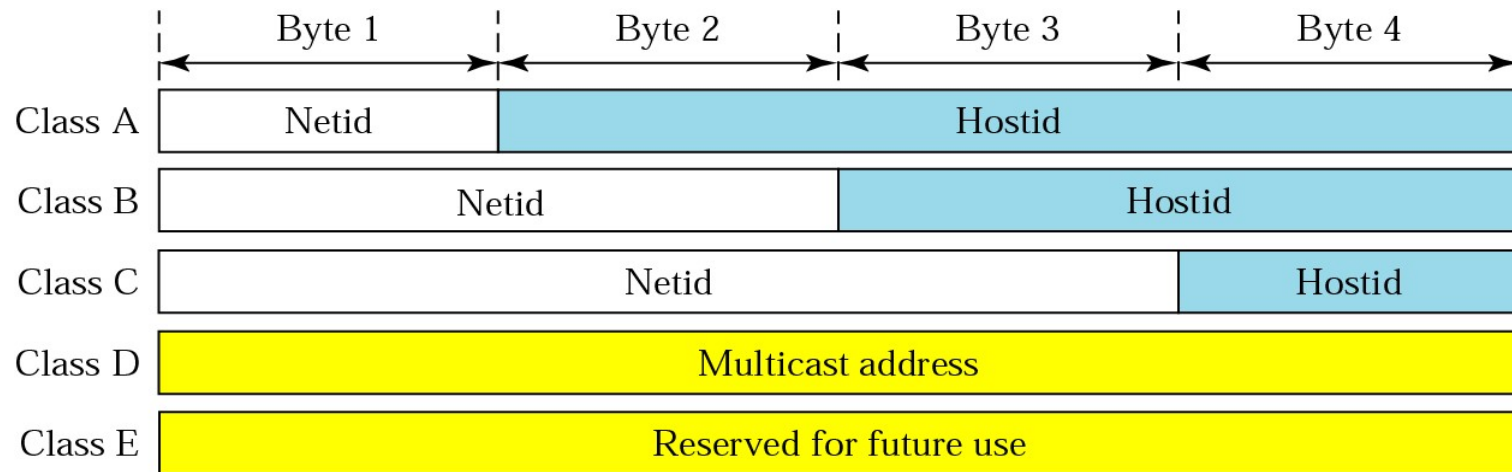


Figure 19.13 Netid and hostid



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Table 19.1 Default masks

Class	<i>In Binary</i>	<i>In Dotted-Decimal</i>	<i>Using Slash</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

SUBNETTING:

Dividing the large network into multiple small NETWORK

192.168.10.0/24

- Class C IP
- /24-network id(24 1's)
- Default subnet mask is 255.255.255.0
- 11111111.11111111.11111111.00000000
- 1-network id; 0- host id

192.168.10.0/25

- Class C IP
- /25-network id(25 1's).
- 11111111.11111111.11111111.10000000(borrow 1 from the next octet)
- Default subnet mask is 255.255.255.128
- 1-network id; 0- host id

Problem 1

- Subnet the ip address into 30 hosts in each subnet

Network Ranges (Subnets)

216.21.5.0 – 216.21.5.31

216.21.5.32 – 216.21.5.63

216.21.5.64 – 216.21.5.95

216.21.5.96 – 216.21.5.127

216.21.5.128 – 216.21.5.159

and so on...

192.168.10.0/25

255								255								255								0							
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

HOW TO FIND THE NUMBER OF NETWORK

2^n (Here, n indicated total number of bits borrowed from host)

$2^1 = 2$ (You can create only two networks)

192.168.10.0/25

255								255								255								0							
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

HOW TO FIND THE NUMBER OF NETWORK

2^n (Here, n indicated total number of bits borrowed from host)

$2^1 = 2$ (You can create only two networks)

HOW TO FIND THE NUMBER OF IP ADDRESS ON EACH NETWORK

2^n (Here, n indicated total number of host bits)

$2^7 = 128$ (On each network you will have 128 IP Address)

192.168.10.0/25

255								255								255								0							
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

HOW TO FIND THE NUMBER OF NETWORK

2^n (Here, n indicated total number of bits borrowed from host)

$2^1 = 2$ (You can create only two networks)

HOW TO FIND THE NUMBER OF IP ADDRESS ON EACH NETWORK

2^n (Here, n indicated total number of host bits)

$2^7 = 128$ (On each network you will have 128 IP Address)

HOW TO FIND THE NUMBER OF HOSTS IN EACH NETWORK

$2^n - 2$ (Here, n indicated total number of remaining host bits)

$2^7 - 2 = 126$ (You will have total 126 Host IP Address on each network)

NOTE: In every network, the first ip address is reserved for the network id and the last ip address is reserved for broadcast id

192.168.10.0/25

255								255								255								0							
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

NETWORK 1

192.168.10.0

Network ID

192.168.10.1



192.168.10.126

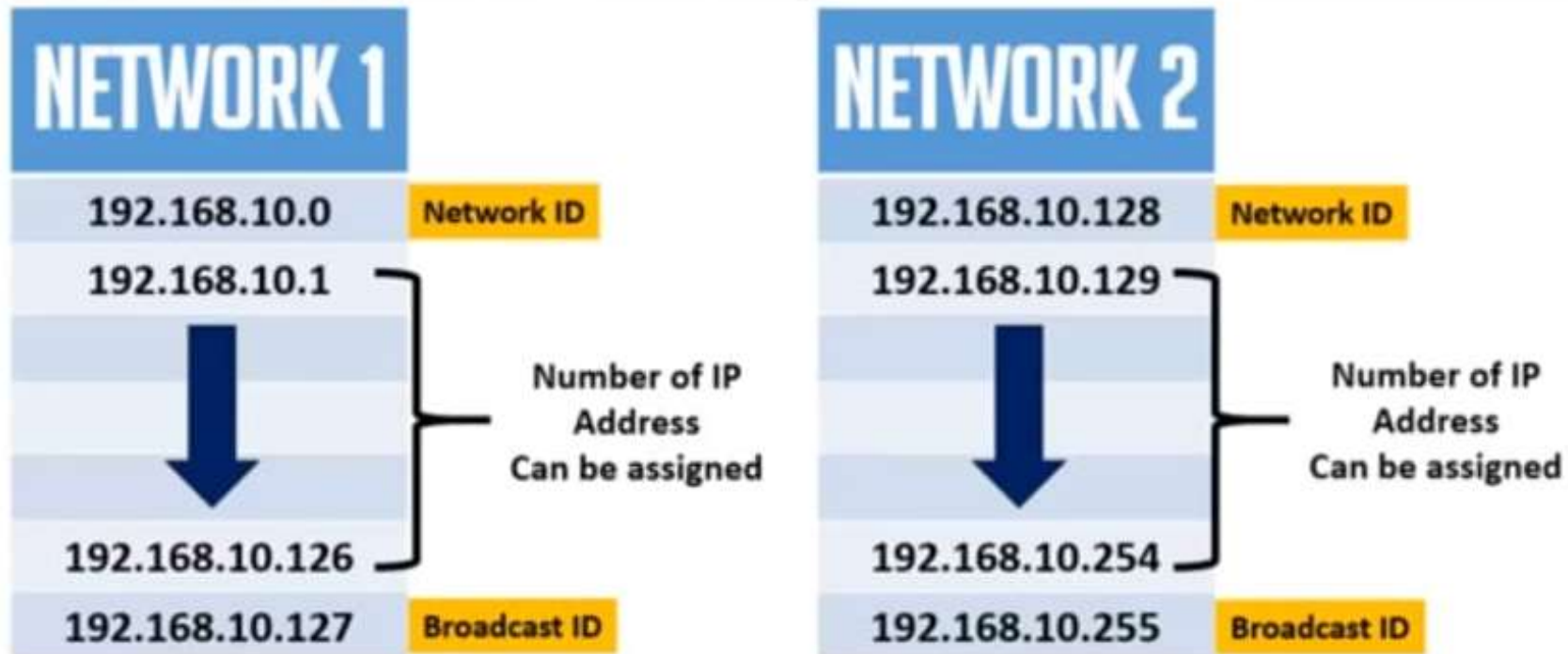
192.168.10.127

Broadcast ID

Number of IP
Address
Can be assigned

192.168.10.0/25

255								255								255								0							
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0



192.168.10.0/26

- Find the no of possible networks
- Find the no of IP address
- Find the no of host on each network

192.168.10.0/26

255								255								255								0							
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Total No of Networks: 4

Total No of IP Address on each network: 64

Total No of Host on each network: 64-2=62

192.168.10.0/26

255								255								255								0							
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Total No of Networks: 4

Total No of IP Address on each network: 64

Total No of Host on each network: 64-2=62

NETWORK NO	NETWORK ID	
1	192.168.10.0	0
2	192.168.10.64	0+64=64
3	192.168.10.128	64+64=128
4	192.168.10.192	128+64=192

192.168.10.0/26

255								255								255								0							
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Total No of Networks: 4

Total No of IP Address on each network: 64

Total No of Host on each network: 64-2=62

NETWORK NO	NETWORK ID	HOST ADDRESS	BROADCAST ID
1	192.168.10.0	192.168.10.1 - 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 -192.168.10.254	192.168.10.255

Fixed Length subnet masking FLSM

What is a fixed-length subnet mask (FLSM)?

- A fixed-length subnet mask (FLSM) refers to a type of enterprise or provider networking where a block of IP addresses is divided into multiple subnets of equal length, i.e. an equal number of IP addresses. FLSM streamlines packet routing within the subnets of a proprietary network.
- A subnet can be a geographically defined local area network (LAN). Alternatively, it may define security boundaries, departmental boundaries, multicast zones or hardware security parameters. One benefit is to create locally significant subnet identification addresses
- Subnetting means dividing a network into multiple smaller subnetworks by using a subnetting mask. In FLSM, the number of IP addresses is the same in each subnet. Another way of saying this is that the same number of IP addresses are allocated to each subnet. Therefore, the subnet mask used will be the same for all the subnets.

What is a Variable-length subnet mask (VLSM)?

- Varying based on required hosts

Problem 1:

An ISP has given a block $10.0.50.0/24$ and organization have 16 departments in which they wish to connect the machines. Find the subnet mask, subnet address of all 16 networks and 1st and last IP of machines in each subnetwork.

Solution:-

Given Network Address:- $10.0.50.0/24$.

Since $/24 \rightarrow$ Subnet mask = $255.255.255.0$.

Total departments needed = $16 = 2^{(4)}$

$\therefore$ For utilization $2^4 = 16 \Rightarrow 4$ bits in host ID has to be borrowed.

$\therefore 24 + 4 \rightarrow (/28) \rightarrow$ New CIDR value.

The first three octet remains constant.

$\therefore$ $11111111 \cdot 11111111 \cdot 11111111 \cdot \boxed{0000}0000$ - 1st Subnet Add
 $11111111 \cdot 11111111 \cdot 11111111 \cdot \boxed{1111}0000$ - 16th Subnet Add

First Subnet block:-

$10.0.50.0/28 \Rightarrow 10.0.50. \boxed{0000}0000$

1st - Subnetwork Address

Possible Address

1st - Broadcast Address

$10.0.50.0$

$10.0.50.1$

$10.0.50.14$

$10.0.50.15$

$10.0.50. \boxed{0000}1111$

Second Subnet block:-

00010000

10.0.50.16/28

$\Rightarrow$ 10.0.50.00010000

2nd Subnet work
Address

$\leftarrow$ 10.0.50.16

Possible Address
to Hosts

10.0.50.17
10.0.50.30

2nd Broadcast
Address

$\leftarrow$ 10.0.50.31
10.0.50.00011111

$$3^{\text{rd}} \text{ Subnet Addr} = 10.0.50.32 / 28 \quad \boxed{0010} 0000$$

$$4^{\text{th}} \text{ Subnet Address} = 10.0.50.48 / 28 \quad \boxed{0011}$$

$$5^{\text{th}} \text{ Subnet Address} = 10.0.50.64 / 28 \quad \boxed{0100}$$

$$6^{\text{th}} \text{ Subnet Address} = 10.0.50.80 / 28 \quad 0101$$

$$7^{\text{th}} \text{ Subnet Address} = 10.0.50.96 / 28$$

$$8^{\text{th}} \text{ Subnet Address} = 10.0.50.112 / 28$$

$$9^{\text{th}} \text{ Subnet Address} = 10.0.50.128 / 28$$

$$10^{\text{th}} \text{ SN Addr} = 10.0.50.144 / 28$$

$$11^{\text{th}} \rightarrow \text{SN Addr} = 10.0.50.160 / 28$$

$$12^{\text{th}} \rightarrow \text{SN Addr} = 10.0.50.176 / 28$$

$$13^{\text{th}} \rightarrow \text{SN Addr} = 10.0.50.192 / 28$$

$$14^{\text{th}} \rightarrow \text{SN Addr} = 10.0.50.208 / 28$$

$$15^{\text{th}} \rightarrow \text{SN Addr} = 10.0.50.224 / 28$$

$$16^{\text{th}} \rightarrow \text{SN Addr} = 10.0.50.240 / 28$$

$$\boxed{1111} 0000$$

Problem 2

An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed length subnets:

- a. Find the subnet mask
- b. Find the number of addresses in each subnet
- c. Find 1st and last address in 1st Subnet
- d. Find 1st and last address in last subnet (500th Subnet)

Subnet Calculator

Number of bits	Number of Subnets
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096

a. Find the subnet mask

An organization is granted the block 16.0.0.0/8

Given Mask: /8 11111111 00000000 00000000 00000000

The administrator wants to create 500 fixed length subnets

So, Number of bits required from host id part is : **9** ($2^9 = 512 \sim 500$)

Subnet mask is: $8 + 9 = 17$

Hence, Subnet mask is /17 or *11111111 11111111 10000000 00000000*

b. Find the number of addresses in each subnet

Subnet mask is /17

Number of bits for Host are: $32-17=15$

Hence, Number of addresses in each subnet = $2^{15} = 32768$

1st Subnet:

1st address: 16.0.0.0 *00010000 00000000 00000000 00000000*

Find 1st and last address in 1st Subnet

1st Subnet:

1st address: 16.0.0.0

00010000 00000000 00000000 00000000

last address: **16.0.127.255**

00010000 00000000 01111111 11111111

For 10th subnet

10th Subnet:

0	0	0	0	0	1	0	1	0
256	128	64	32	16	8	4	2	1

Subnet mask: 11111111 11111111 10000000 00000000

10th subnet address: 00010000 00000101 00000000 00000000

1st address: 16. 5. 0. 0

last address: 16. 5. 127. 255

For 500th
subnet

d. Find 1st and last address in last subnet (500th Subnet)

500th Subnet:

1	1	1	1	1	0	0	1	1
256	128	64	32	16	8	4	2	1

Subnet mask: 11111111 11111111 10000000 00000000

500th subnet address: 00010000 11111001 10000000 00000000

1st address: 16. 249. 128. 0

last address: 16. 249. 255. 255

500th Subnet address: 16.249.128.0/17

1st address: 16.249.128.0

Last Host Address: 16.249.255.254

Problem 3

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

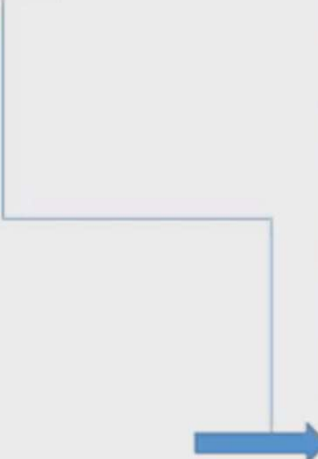
- a. The first group has 64 customers; each needs 256 addresses.*
- b. The second group has 128 customers; each needs 128 addresses*
- c. The third group has 128 customers; each needs 64 addresses.*

Number of Host (Theoretically), practically $2^n - 2$

Number of bits (Suffix)	Number of Host
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096

Given block is: 190.100.0.0/16

*The first group has 64 customers; each needs **256** addresses.*



Number of bits (Suffix)	Number of Host
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096

Given block is: 190.100.0.0/16

The first group has 64 customers; each needs 256 addresses.

Mask is $32-8=24$

Customer-1: 190.100.0.0/24 to 190.100.0.255/24

Customer-2: 190.100.1.0/24 to 190.100.1.255/24

Customer-64: 190.100.63.0/24 to 190.100.63.255/24

Number of bits (Suffix)	Number of Host
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096

Given block is: 190.100.0.0/16

The second group has 128 customers; each needs 128 addresses.

Mask is $32 - 7 = 25$

Available after Group-1 last customer:

Customer-1: 190.100.64.0/25 to 190.100.64.127/25

Customer-2: 190.100.64.128/25 to 190.100.64.255/25

Customer-3: 190.100.65.0/25 to 190.100.65.127/25

Customer-128: 190.100.127.128/25 to 190.100.127.255/25

Number of bits (Suffix)	Number of Host
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096

ISP

Granted Addresses:

190.100.0.0

to

190.100.255.255

Group 1:
190.100.0.0 to 190.100.63.255

Customer 001: 190.100.0.0/24

⋮

Customer 064: 190.100.63.0/24

Group 2:
190.100.64.0 to 190.100.127.255

Customer 001: 190.100.64.0/25

⋮

Customer 128: 190.100.127.128/25

Group 3:
190.100.128.0 to 190.100.159.255

Customer 001: 190.100.128.0/26

⋮

Customer 128: 190.100.159.192/26

Available
190.100.160.0 to 190.100.255.255